INTERNATIONAL MONETARY FUND

MALI

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SECOND AND THIRD REVIEWS UNDER THE EXTENDED CREDIT FACILITY ARRANGEMENT, REQUESTS FOR WAIVERS OF NONOBSERVANCE OF PERFORMANCE CRITERIA AND MODIFICATION OF PERFORMANCE CRITERION—DEBT SUSTAINABILITY ANALYSIS

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Mali: Joint Bank-Fund Staff Debt Sustainability Analysis	
Risk of external debt distress:	*Moderate*
Overall risk of debt distress	*Moderate*
Granularity in risk rating	*Moderate risk tool: Some space to absorb shocks*
Application of judgement	*Yes, customized scenario is considered for external and overall debt distress risk rating*

Mali remains at moderate risk of both external and overall debt distress consistent with the joint Bank-Fund assessment from May 2020, but vulnerabilities have increased. The pandemic and coup-induced deterioration in the fiscal balance, a more gradual consolidation path towards the WAEMU fiscal target, and a shortfall in external donor grant support and thus a heavier reliance on more expensive domestic borrowing are driving a substantial increase in public debt levels and gross financing needs. Standardized stress tests do not result in breaches of DSA thresholds for either external or public debt, but a customized scenario demonstrates the vulnerability of Mali's external debt to a change in security conditions or other fiscal pressures that could lead to larger fiscal deficits financed on non-concessional terms.[1] Under this customized scenario, Mali's public debt and debt service levels would breach their respective thresholds if commodity prices and exports were to deteriorate, highlighting sustainability and liquidity risks. To avoid a deterioration in debt levels and the debt servicing burden it is essential that the authorities continue their efforts to mobilize tax revenue and implement structural reforms that address fiscal risks from state-owned enterprises and the cost of the civil service, as well as strengthen debt and public financial management.

[1] This DSA was jointly prepared by IMF and World Bank staff under the debt sustainability framework (DSF) for low-income countries (LICs), implemented since July 2018. The debt-carrying capacity is classified using the country-specific composite indicator (CI) composed of three macroeconomic indicators and the World Bank's Country Policy and Institutional Assessment (CPIA). Mali's capacity is assessed as "medium" using the CI (with a CI index of 2.90) based on the 2020 October WEO and 2019 CPIA that was released in July 2020.

PUBLIC DEBT COVERAGE

1. Mali's public debt covers external and domestic obligations of the central government (Text Table 1). State and local government entities do not borrow directly on their own. Detailed information on SOE debt is currently not available; the SOE contingent liability shock has be therefore been revised upward to 5 percent of GDP from the default shock of 2 percent. Staffs are supporting authorities' efforts to broaden the coverage of public debt to include SOE debt and debt of other public institutions outside the central government in public debt reports. Improvements in debt recording and reporting are critical. As for external debt, it is classified based on the currency denomination.[2]

Text Table 1. Coverage of Public Debt and Size of Contingent Liability Shock

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

2. A contingent liability test with tailored magnitude of shocks is applied to reflect the potential impacts of additional liabilities (Text Table 2). The component of the contingent liability shock related to SOEs that are not accounted for in the public sector coverage is estimated to be at 5 percent of GDP, 3 percentage points higher than the default setting. Public Private Partnerships (PPPs) capital stock for Mali is estimated at 3.7 percent of GDP from the World Bank's PPP database, implying a calibrated 1.3 percent of GDP for the PPP component of the contingent liability stress test. The shock emanated from the financial sector is set at the default level of 5 percent of GDP. Total contingent liabilities for the stress test are estimated at 11.3 percent of GDP.

Text Table 2. Size of Contingent Liability Shock

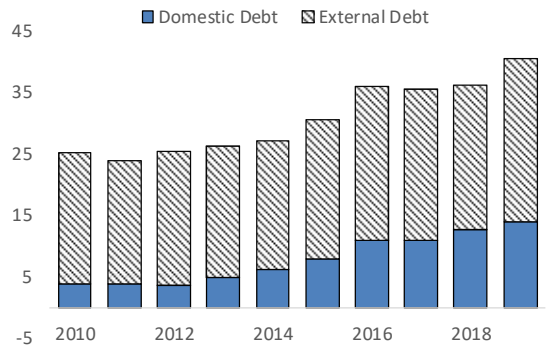
1	**The country's coverage of public debt**	The central government, government-guaranteed debt	
		Default	**Used for the analysis**
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.0
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	5.0
4	PPP	35 percent of PPP stock	1.3
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0
	Total (2+3+4+5) (in percent of GDP)		11.3

[2] BOAD (West African Development Bank) loans are treated as external debt, although they are denominated in CFAF, because program targets were set when BOAD loans were classified as external debt.

DEBT BACKGROUND

3. At end–2019, Mali's stock of public debt amounted to 40.5 percent of GDP, with almost two-thirds consisting of external debt (Text Figure 1 and Text Table 3). External debt amounted to CFAF 2,682 billion (26.4 percent of GDP), of which CFAF 2,044 billion (20.1 percent of GDP) to multilateral creditors and CFAF 638 billion (5.3 percent of GDP) to bilateral creditors. The main external creditors—accounting for 90 percent of the external debt stock—include IMF, the World Bank, African Development Fund, West African Development Bank, Islamic Development Bank, France (French Development Agency), China, India and Abu Dhabi (Abu Dhabi Development Fund). However, domestic debt has rapidly built up, increasing from a low base of 6.3 percent of GDP in 2014 to 14.0 percent of GDP in 2019, mostly in treasury bills and bonds issued on the WAEMU regional market. The increasing share of domestic debt increases the debt service burden on Mali, due to its higher interest rates (an average effective rate of 5.8 percent compared to 1.4 percent for external debt).

Text Figure 1. Mali: Evolution of Public Debt 2010-2019
(in percent of GDP)



Source: Malian authorities.

Text Table 3. Mali: External Debt Stock at Year-End, 2014–2019
(billions of CFAF)

	2014	2015	2016	2017	2018	2019
Total	1,485	1,754	2,074	2,182	2,215	2,682
(percent of GDP)	20.9	22.6	25.0	24.5	23.4	26.4
Multilateral	1,202	1,384	1,626	1,754	1,755	2,044
IMF	94	80	98	69	48	173
World Bank/IDA	655	814	849	872	969	1,163
African Development Fund	245	253	343	387	398	410
West African Development Bank	57	66	65	122	108	122
Islamic Development Bank	92	116	141	118	99	98
Others	59	56	131	186	133	79
Official Bilateral	282	370	448	429	460	638
Paris Club	16	49	50	48	51	87
France	3	34	34	34	38	73
Korea	7	8	8	8	8	8
Austria	2	3	3	3	3	3
Belgium	4	4	4	3	2	2
Non-Paris Club	266	321	398	380	409	551
China	149	193	263	262	294	249
India	78	84	83	71	75	58
Abu Dhabi	0	0	0	0	0	165
Others	266	321	398	380	409	386

Sources: Malian authorities.

4. The pandemic-induced economic recession and widening fiscal deficit as well as other fiscal pressures are leading to a notable increase in total debt in 2020. The overall deficit for 2020 is projected to widen to 5.5 percent of GDP compared to 3.5 percent of GDP targeted at the time of the first ECF review. As a result, public debt is projected to increase to 45.9 percent of GDP at end-2020. Additionally, given the reduction in budget support from international donors following the coup, the fiscal deficit is projected to be largely financed by domestic borrowing, which increases the share of domestic debt in total public debt to a record high at around 40 percent. The authorities were temporarily unable to service debt due to post-coup sanctions in place between late August and early October 2020.[3]

5. To cope with the fallout from the pandemic, Mali has been granted debt relief under the Catastrophe Containment and Relief Trust (CCRT) and received debt service suspension under the Debt Service Suspension Initiative (DSSI). These initiatives are expected to lower its debt payments in 2020–22. The CCRT is provided as a grant covering eligible debt payments due to the IMF. The 1st and 2nd tranches have provided relief on debt payments due to the IMF during April 2020-April 2021 period of CFAF 11.5 billion (about 0.12 percent of GDP). The 3rd and 4th tranches are expected to provide relief on debt payments of CFAF 12.2 billion (about 0.12 percent of GDP) due during April 2021-April 2022 period. The authorities also requested debt relief under the G20 DSSI initiative in 2020, which aims to ease financing constraints and free up money to respond to the pandemic. The authorities formally requested relief from the Paris Club and other official bilateral creditors for debt service amounting to 0.25 percent of GDP during May-December 2020 period, of which only 0.07 percent of GDP was effectively suspended by France, Belgium, Korea, Saudi Arabia, Libya and Kuwait. The authorities also formally requested relief under the DSSI extension in 2021, with an expected reduction in debt service of 0.23 percent of GDP for the January-June 2021 period, and have received agreement on debt service suspension of 0.05 percent of GDP (Text Table 4).

Text Table 4. Mali: DSSI Delivery and Projection
(millions of USD)

	2020 (May-December)			2021 (January-June)		
	Eligible	Requested	Effective	Eligible	Requested	Effective
Paris Club	**3.6**	**3.6**	**3.6**	**9.0**	**9.0**	**9.0**
France	3.1	3.1	3.1	9.0	9.0	9.0
Belgium	0.5	0.5	0.5	-	-	-
Korea	0.0	0.0	0.0	0.0	0.0	0.0
Non-Paris Club	**26.1**	**40.8**	**9.1**	**35.8**	**35.8**	-
G20						
China	15.6	15.6	-	19.1	19.1	Waiting
India	7.7	7.7	-	10.0	10.0	for
Saudi Arabia	2.8	2.8	2.8	6.8	6.8	response
Others						
Libya	-	6.5	3.2	-	Request not sent	
Kuwait	-	4.2	3.1	-	Request not sent	
Iraq	-	2.0	-	-	Request not sent	
Abu Dhabi	-	1.9	-	-	Request not sent	
Total	29.7	44.3	12.7	44.8	44.8	9.0
(percent of GDP)	0.17	0.25	0.07	0.23	0.23	0.05

Sources: Malian authorities.

[3] During this period, total missed debt service payments on regional and external debt amounted to US$57 million or 0.3 percent of GDP, of which 0.2 of a percentage points to external creditors, including the World Bank, syndicated debt, OPEC, Saudi Funds for Development and Exim Bank of China, and 0.1 of a percentage point to regional creditors. Of the missed payments, only 0.01 percent of GDP represented arrears as other payments were settled within respective grace periods (after sanctions were lifted and the government regained access to the Treasury Single Account at the BCEAO in early October 2020), including on IDA debt service.

UNDERLYING MACROECONOMIC ASSUMPTIONS AND COUNTRY CLASSIFICATION

A. Background on Macroeconomic Forecasts

6. This DSA is consistent with the macroeconomic framework underlying the staff report for the second and third review under the ECF arrangement. Macroeconomic conditions have been updated from the previous DSA (Text Table 5). In the near term, the main assumptions have changed as a result of the pandemic, the coup and heightened fiscal pressures. Changes are reflected in growth, inflation, fiscal balance, trade growth, current account, and external support. In the medium and long term, key macroeconomic assumptions are in line with the previous DSA.

- **Lower GDP growth in 2020 but a rebound from 2021 onward.** The outbreak of the COVID-19 pandemic and the coup d'état in late August have disrupted a prolonged period of strong economic performance. Economic growth is estimated to fall to around -2 percent in 2020 and expected to pick up in 2021 to 4 percent, before settling at its estimated potential growth of 5 percent over the medium term (Text Table 5). Despite the projected strong economic recovery, downside risks dominate, stemming especially from the prospect of continued political instability, a challenging security environment, heightened fiscal pressures, and uncertainties around the dynamics of the pandemic. The double shock from the pandemic and the coup contributed to a deteriorated outlook in 2020 from the previous DSA (-0.9 percent versus 2 percent growth), while not affecting it going forward.[4]
- **A re-anchored more gradual path for fiscal consolidation.** Fiscal performance was strong in 2019, with the overall deficit narrowing to 1.7 percent of GDP compared to the 3 percent of GDP ceiling under the program. In 2020, the overall deficit is projected to widen to 5.5 percent of GDP, with many factors contributing to the deterioration including lower revenues and higher spending due to the pandemic, wage increases in the education sector, and tighter donor grant support post-coup. While most of the COVID-19 related measures are expected to be rolled back in 2021 and international partners have started to re-engage with Mali, a number of factors are still expected to weaken the medium-term fiscal position, including the structural increase in the wage bill (1.3 percent of GDP), the weak position of state-owned enterprises and a lower yield from tax administration measures. In light of these pressures and the weaker cyclical position, the revised program envisages a more gradual return of the fiscal deficit to the regional WAEMU ceiling of 3 percent of GDP by 2024. The gradual path of the fiscal consolidation takes into account of the authorities' continued commitment to priority social spending, which is reflected in the indicative targets in 2020 and 2021.
- **Narrowed current account deficit in the near term.** The current account deficit is projected to narrow to 2.0 percent of GDP in 2020 (3.6 in previous DSA) from 4.8 percent of GDP in 2019, driven by significantly higher gold prices (main export) and lower oil prices (main import). Going

[4] A full DSA was done when the RCF was approved in April 2020 (IMF Country Report 20/153), but without a full writeup which was temporarily waived in light of the pandemic.

forward, the current account will widen over the medium term with stronger domestic demand and the gradual deterioration in the terms of trade (although it is expected to remain notably above historical levels). The current account deficit is projected to be financed mainly through official grants and loans, FDI, and IMF programs.

Text Table 5. Mali: Evolution of Selected Projected Macroeconomic Indicators

	2017	2018	2019	2020	2021	2022	2023	2024	Long term[1]
				Projections					
Real GDP growth									
Current DSA	5.3	4.7	4.8	-2.0	4.0	6.0	5.0	5.0	5.0
Previous DSA	5.0	5.2	5.1	0.9	4.0	6.0	5.0	5.0	5.0
GDP Deflator									
Current DSA	4.0	6.2	-3.3	3.7	6.7	2.8	2.4	2.2	2.9
Previous DSA	4.2	5.1	-2.9	1.3	2.8	2.5	2.3	2.2	2.2
Overall fiscal balance (including grants, percent of GDP)									
Current DSA	-2.9	-4.7	-1.7	-5.5	-5.5	-4.5	-3.5	-3.0	-3.0
Previous DSA	-2.9	-4.8	-1.7	-6.2	-3.0	-3.0	-3.0	-3.0	-3.0
Export of goods and services growth (percent)									
Current DSA	5.0	21.3	3.1	10.8	7.0	3.7	2.1	2.3	4.1
Previous DSA	5.0	21.3	1.8	1.5	11.0	5.0	5.9	6.1	4.5
Import of goods and services growth (percent)									
Current DSA	-1.7	9.5	2.5	-9.0	13.8	8.4	5.9	6.0	6.4
Previous DSA	-1.7	9.5	0.3	-8.4	12.0	9.3	7.0	6.2	6.9
Current account deficit (including grants, percent of GDP)									
Current DSA	-7.3	-4.9	-4.8	-2.0	-2.4	-2.9	-3.6	-4.5	-8.5
Previous DSA	-7.3	-4.9	-4.2	-3.6	-3.8	-5.4	-6.5	-6.7	-17.0
Official aid[2] (percent of GDP)									
Current DSA	3.9	2.3	5.6	2.9	3.7	5.0	5.2	4.9	4.5
Previous DSA	3.9	2.3	5.6	5.4	4.4	3.8	3.7	3.7	3.2

Sources: Malian authorities; and IMF Staff estimates and projections.

[1] Defined as the last 15 years of the projection period. For the current DSA, the long term covers the 2025-39 period. The previous DSA refers to the full DSA in the RCF request in May 2020 (IMF Country Report 20/153), for which the long term covers the same period from 2025 to 2039.

[2] Defined as the sum of concessional grants and loans.

- **A shortfall in external support in the near term.** Gross financing needs are projected to increase and peak in 2022, mainly driven by larger targeted deficit and heavier reliance on domestic financing that has shorter maturities and higher interest rates (compared to external financing). External borrowing (including from the Fund) is estimated to have decreased to 2.9 percent of GDP in 2020 from 3.9 percent of GDP in 2019, reflecting the temporary disengagement of international and regional partners after the coup (Text Table 6). This has created financing challenges for the execution of the 2020 budget that may persist into 2021. In particular, since budget support from the World Bank has shifted to project investment in 2020-21 and other donors are still firming up their budget support, the baseline scenario assumes lower budget support in 2021, somewhat offset by higher project lending. Over the medium term, gross financing needs will continue to be met by external multilateral sources and bonds issued in the domestic and regional bond market (UMOA-Titres and APA).

Text Table 6. Mali: External Financing, 2019-2021
(billions of CFAF)

	2019	2020	2021
		Est.	Proj.
Loans	377	157	192
(percent of GDP)	3.7	1.5	1.8
Project loans, o/w:	105	95	189
World Bank	37	25	69
Budgetary loans, o/w:	271	62	3
World Bank	124	0	0
Grants	191	141	202
(percent of GDP)	1.9	1.4	1.9
Project grants, o/w:	79	75	156
World Bank	26	28	51
Budget grants, o/w:	112	66	46
World Bank	24	0	0
IMF (CCRT)	-	12	12
IMF	16	134	63
(percent of GDP)	0.2	1.3	0.6
ECF	16	17	63
RCF	-	117	-

Sources: Malian authorities and IMF staff estimates and projections.

7. The realism tools show that projections are broadly in line with historical and peers' experiences (Figures 3 and 4).

- **Debt dynamics**. The dynamics of public debt shows a deterioration in the near to medium term because of the wider fiscal deficit and the more gradual return to WAEMU target, while that of external debt shows an improvement compared to the previous DSA in the near term. This different dynamic is the result of the projected shortfall in external financing in 2020 and 2021, and therefore a higher reliance on domestic borrowing and a heavier debt service burden. The projected contributions of different debt-creating factors do not deviate significantly from the historical outturns, except for the negative contribution of price and exchange rate in the projection mainly driven by the appreciation path of CFAF.
- **Fiscal adjustment**. The projected 3-year change in the primary balance (-2.1 percent of GDP) lies below the 75th percentile of the distribution of approved Fund-supported programs for LICs since 1990s, suggesting the adjustment in the primary fiscal balance is in line with peer country experiences. The difference between the baseline and the expected fiscal impulse in 2020 is explained by the economic recession caused by the pandemic and the necessary policy response to it.
- **Public investment and growth.** The public investment to GDP ratio is projected to decline over the medium term, resulting in a lower contribution of government capital stock to growth (24

percent versus 30 percent in the previous DSA, and 44 percent at the historical level), largely driven by the expenditure switch to a higher wage bill. However, the baseline scenario does not yet include the anticipated increase in revenues from the tax package to be finalized ahead of the 2022 budget; if tax reforms are carried through, additional fiscal space will open for public investment.

COUNTRY CLASSIFICATION AND DETERMINATION OF STRESS TESTS

8. Mali is assessed as having a *medium* debt carrying capacity. Based on the October 2020 WEO and updated World Bank's 2019 CPIA measures, Mali's composite indicator score is calculated to be 2.90 (above the lower cut-off of 2.69 but below the strong capacity cut-off of 3.05), in line with the assessment of medium debt carrying capacity applied in the previous DSA with a slight decrease mainly because of lower world economic growth. The relevant thresholds are used to assess external debt risk rating.

Text Table 6. Mali: Debt Carrying Capacity and CI Score

Debt Carrying Capacity	Medium		
Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Medium	Medium 2.90	Medium 3.01	Medium 3.00

Note: Until the April 2019 WEO vintage is released, the two previous vintages ago classification and corresponding score are based solely on the CPIA per the previous framework.

Calculation of the CI Index

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.396	1.31	45%
Real growth rate (in percent)	2.719	4.586	0.12	4%
Import coverage of reserves (in	4.052	39.093	1.58	55%
Import coverage of reserves^2 (in percent)	-3.990	15.283	-0.61	-21%
Remittances (in percent)	2.022	5.051	0.10	4%
World economic growth (in percent)	13.520	2.928	0.40	14%
CI Score			2.90	100%
CI rating			Medium	

9. A customized scenario is applied in addition to standardized stress tests and tailored test on commodity price. In light of the downside risks stemming from the prospect of continued political instability, a challenging security environment, heightened fiscal pressures, and uncertainties surrounding the dynamics of the pandemic, the customized scenario illustrates risk from a deterioration in the primary fiscal balance relative to the baseline (2.5 percent of GDP higher in 2021, and then 2 percent of GDP higher

in 2022 and 2023), which generates financing gaps that are assumed to be closed by more expensive external borrowing due to a fixed envelope of available concessional financing and grants.

EXTERNAL DEBT SUSTAINABILITY ANALYSIS

10. External debt is projected to decline in the near term but increase after 2023 as external official support resumes. Under the baseline scenario, external debt is expected to decline in the near term due to the shortfall in external official support during 2020-21 and to reach 25.6 percent of GDP in 2023 (Table 1). As external official budget support is assumed to resume at its historical level starting in 2023, external debt is projected to be on an upward path until around 2030, reaching 28.5 percent of GDP, before settling at around 28 percent of GDP at the end of the projection period in 2040.

11. The risk of external debt distress remains moderate with some space to absorb shocks.

- All external debt indicators remain below their corresponding indicative thresholds under the baseline scenario and standardized stress tests, indicating space to absorb shocks. Under the baseline scenario, the PV of external debt-to-GDP ratio is projected to decrease from 21.4 percent in 2020 to 20.1 percent in 2025, and then increase back to 21.3 percent in 2030, consistent with the nominal external debt stock dynamics (Figure 1 and Table 3), below the indicative threshold of 40 percent throughout the projection period. The PV of external debt-to-exports ratio, although on an upward trajectory mainly driven by declining export in the medium to long term, is also comfortably below the 180 percent threshold. Debt service indicators are all stable and below their corresponding thresholds.
- However, under a customized fragility scenario of a deterioration in the fiscal position as a result of any of the risks discussed above, shock to export growth would result in two breaches of the 15 percent threshold for external debt service-to-exports (by 0.2 of a percentage point in 2024 and 0.4 of a percentage point in 2027), downgrading the risk rating from low to moderate. The space to absorb a depreciation shock in the customized scenario is not substantial either, with debt service-to-revenue ratio almost breaching the 18 percent threshold in 2024.

PUBLIC DEBT SUSTAINABILITY ANALYSIS

12. Public debt is projected to be on an upward trend in the near term, mainly driven by higher deficits financed by more domestic borrowing. Under the baseline scenario, public debt is expected to increase to 47 percent of GDP in 2023, and then gradually decline, reaching 43.2 percent of GDP in 2030 driven by the favorable automatic debt dynamics and the return of fiscal deficit to the WEAMU ceiling (Table 2). In the very long run, public debt is expected to stabilize at below 40 percent of GDP at the end of the projection period in 2040. The share of domestic debt in public debt follows a similar trend, peaking at 45.5 percent in 2022 (almost 10 percentage points higher than that in 2019), and declining gradually afterwards (reaching 34.1 percent in 2030), close to historical levels.

13. The risk of public debt distress also remains moderate. Consistent with the public debt dynamics, under the baseline scenario the PV of public debt-to-GDP ratio increases to about 41 percent in 2022-23 and then declines to 36 percent in 2030, remaining under the 55 percent indicative threshold

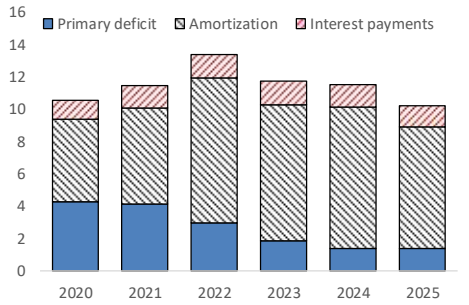
throughout the projection period, even under shocks from stress tests (Figure 2 and Table 4). Other debt and debt service indicators also follow similar trends. However, under the customized scenario discussed above (a market-financed deterioration in the fiscal balance), the indicative thresholds for public debt sustainability are also breached from 2023 onwards (by 0.6 of a percentage point in 2023 and almost 7 percentage points in 2030)—in this case under a shock to commodity prices, which would lower growth and revenues—bringing the risk rating for overall public debt to moderate.

RISK RATINGS AND VULNERABILITIES

14. Mali's risk of external debt distress and overall debt distress remain moderate. While none of the external debt or public debt indicators breach their corresponding thresholds under the baseline scenario, a customized scenario of higher deficits financed at unfavorable terms underscore the vulnerabilities of Mali's external and overall debt to shocks. In such case, a shock to the commodity price would result in public debt being on a firm upward trajectory and breaching the threshold in only three years. The projected shortfall in external official support in 2020-21 improve external debt dynamics in the near term, however, liquidity risks associated with a higher debt service burden are captured in the customized scenario, in which a shock to export growth would result in a breach of the indicative threshold for the external debt service indicator.

15. Financing risks are increasing and need to be closely monitored. Larger primary deficits, tighter donor support, at least in the near term, and increasing debt service burden are compounding financing risks. Gross financing needs are estimated to average 12 percent of GDP over the next two years and 11.5 percent of GDP over the next five years, straining the authorities' ability to meet them (Text Figure 2). The need to tap the regional markets as the main source of financing in the near term will add risks not just through more expensive borrowing but also due to potential market tightening if other WAEMU countries finance their larger deficits in the regional market at the same time. Hence, Mali needs to maintain prudent macroeconomic policies to ensure macroeconomics stability, and proactively engage with external official partners to secure more grants and concessional loans, while strengthening the effectiveness of debt management and tax revenue mobilization. In addition, structural reforms that prioritize areas of fiscal pressure should continue and deepen, including measures to prevent arrears accumulation, manage the wage bill, and SOE performance to safeguard the fiscal space for needed developmental, social and security spending.

Text Figure 2. Mali: Gross Financing Needs 2020-2025
(in percent of GDP)



Source: Malian authorities.

AUTHORITIES' VIEWS

16. The authorities have agreed that Mali remains at moderate risk of debt distress with some room to absorb shocks. They have concurred with the importance of external support from international donors with whom they have been actively re-engaging. They have stressed that their debt strategy aims at

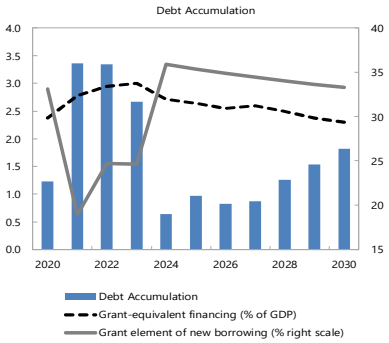
mobilizing more concessional external loans, favoring domestic loans with longer maturities, maintaining a prudent policy to mobilize fixed-rate loans, and implementing a more active policy to choose currencies of loans (although flexibility is not very large).

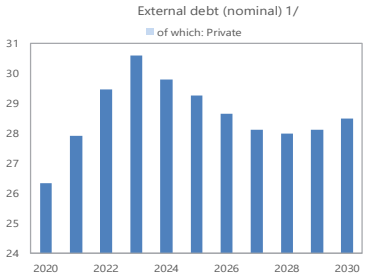
Table 1. Mali: External Debt Sustainability Framework, Baseline Scenario, 2017–40

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
External debt (nominal) 1/	**24.5**	**23.4**	**26.4**	**26.3**	**27.9**	**29.5**	**30.6**	**29.8**	**29.3**	**28.5**	**28.2**	**22.8**	**28.6**
of which: public and publicly guaranteed (PPG)	**24.5**	**23.4**	**26.4**	**26.3**	**27.9**	**29.5**	**30.6**	**29.8**	**29.3**	**28.5**	**28.2**	**22.8**	**28.6**
Change in external debt	-0.5	-1.1	3.1	-0.1	1.6	1.5	1.1	-0.8	-0.5	0.4	-1.1		
Identified net debt-creating flows	**1.5**	**-0.3**	**2.8**	**1.1**	**0.3**	**0.2**	**1.1**	**1.9**	**2.8**	**5.8**	**7.6**	**1.8**	**2.8**
Non-interest current account deficit	**7.0**	**4.7**	**4.5**	**1.6**	**2.1**	**2.3**	**2.9**	**3.7**	**4.7**	**7.8**	**9.6**	**5.4**	**4.6**
Deficit in balance of goods and services	13.7	11.1	11.1	5.1	6.8	7.9	8.7	9.4	10.0	11.9	12.5	**12.7**	**9.4**
Exports	22.5	24.5	24.9	27.2	26.2	25.0	23.7	22.6	21.7	18.0	12.5		
Imports	36.2	35.6	36.0	32.2	33.1	32.9	32.4	32.0	31.7	29.9	25.0		
Net current transfers (negative = inflow)	-9.7	-9.2	-9.3	-6.6	-7.9	-8.6	-8.6	-8.4	-8.2	-7.3	-6.0	**-10.3**	**-7.9**
of which: official	-4.8	-4.4	-4.8	-2.3	-3.3	-3.9	-3.8	-3.5	-3.2	-2.3	-1.0		
Other current account flows (negative = net inflow)	3.0	2.8	2.7	3.1	3.1	3.0	2.9	2.8	3.0	3.1	3.1	**3.0**	**3.0**
Net FDI (negative = inflow)	**-3.6**	**-2.7**	**-1.8**	**-1.4**	**-1.1**	**-1.1**	**-1.1**	**-1.1**	**-1.1**	**-1.1**	**-1.1**	**-2.6**	**-1.2**
Endogenous debt dynamics 2/	**-1.9**	**-2.2**	**0.0**	**0.9**	**-0.6**	**-1.0**	**-0.7**	**-0.7**	**-0.8**	**-0.9**	**-0.9**		
Contribution from nominal interest rate	0.3	0.2	0.3	0.4	0.4	0.5	0.7	0.7	0.6	0.4	0.5		
Contribution from real GDP growth	-1.2	-1.0	-1.1	0.5	-1.0	-1.5	-1.4	-1.4	-1.4	-1.3	-1.4		
Contribution from price and exchange rate changes	-1.0	-1.4	0.8	...	...	...	...	...	...	...	...		
Residual 3/	**-2.0**	**-0.8**	**0.3**	**-1.2**	**1.2**	**1.4**	**0.0**	**-2.7**	**-3.3**	**-5.4**	**-8.7**	**-1.2**	**-2.6**
of which: exceptional financing	-0.2	-0.2	0.0	0.0	0.0	-0.1	-0.1	-0.1	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**20.5**	**21.4**	**22.3**	**23.5**	**24.4**	**23.3**	**22.9**	**21.3**	**21.8**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**82.3**	**78.7**	**85.1**	**94.3**	**102.9**	**103.1**	**105.3**	**118.6**	**174.6**		
PPG debt service-to-exports ratio	**4.6**	**3.5**	**5.0**	**6.6**	**6.7**	**8.0**	**10.6**	**12.6**	**11.6**	**8.6**	**12.9**		
PPG debt service-to-revenue ratio	**5.6**	**6.0**	**6.3**	**9.7**	**10.2**	**11.2**	**13.8**	**14.4**	**12.6**	**7.7**	**10.4**		
Gross external financing need (Million of U.S. dollars)	691.2	479.0	689.2	347.8	525.6	678.3	985.4	1337.6	1593.1	3048.3	8463.9		
Key macroeconomic assumptions													
Real GDP growth (in percent)	5.3	4.7	4.8	-2.0	4.0	6.0	5.0	5.0	5.0	5.0	5.0	**4.4**	**4.4**
GDP deflator in US dollar terms (change in percent)	4.0	6.2	-3.3	3.7	6.7	2.8	2.4	2.2	1.1	2.0	2.7	**1.3**	**2.7**
Effective interest rate (percent) 4/	1.1	1.0	1.4	1.5	1.6	2.1	2.4	2.6	2.1	1.5	1.7	**0.6**	**1.9**
Growth of exports of G&S (US dollar terms, in percent)	5.0	21.3	3.1	10.8	7.0	3.7	2.1	2.3	2.1	3.4	4.1	**6.7**	**4.1**
Growth of imports of G&S (US dollar terms, in percent)	-1.7	9.5	2.5	-9.0	13.8	8.4	5.9	6.0	5.3	6.0	4.9	**6.6**	**5.5**
Grant element of new public sector borrowing (in percent)	...	...	...	33.1	19.0	24.7	24.7	35.9	35.4	33.3	31.8	**...**	**31.2**
Government revenues (excluding grants, in percent of GDP)	18.4	14.3	19.5	18.6	17.4	17.8	18.1	19.8	19.9	20.2	15.4	**15.8**	**19.3**
Aid flows (in Million of US dollars) 5/	248.1	210.8	326.5	368.8	621.8	726.5	785.7	756.8	768.8	867.2	1150.7		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	2.4	2.8	2.9	3.0	2.7	2.6	2.3	1.5	**...**	**2.6**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	54.3	42.0	42.6	44.2	55.5	54.4	50.4	49.7	**...**	**50.4**
Nominal GDP (Million of US dollars)	15,360	17,079	17,308	17,592	19,519	21,274	22,880	24,560	26,074	37,108	84,055		
Nominal dollar GDP growth	9.5	11.2	1.3	1.6	11.0	9.0	7.5	7.3	6.2	7.1	7.8	**5.7**	**7.2**
Memorandum items:													
PV of external debt 7/	...	...	20.5	21.4	22.3	23.5	24.4	23.3	22.9	21.3	21.8		
In percent of exports	...	...	82.3	78.7	85.1	94.3	102.9	103.1	105.3	118.6	174.6		
Total external debt service-to-exports ratio	4.6	3.5	5.0	6.6	6.7	8.0	10.6	12.6	11.6	8.6	12.9		
PV of PPG external debt (in Million of US dollars)			3553.0	3765.6	4356.6	5008.1	5574.6	5720.2	5958.0	7922.6	18312.8		
(PVt-PVt-1)/GDPt-1 (in percent)				1.2	3.4	3.3	2.7	0.6	1.0	1.8	1.0		
Non-interest current account deficit that stabilizes debt ratio	7.5	5.8	1.4	1.7	0.5	0.8	1.8	4.5	5.3	7.4	10.7		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

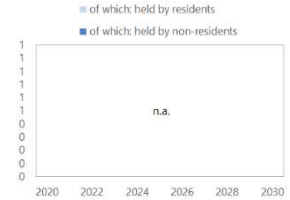
Table 2. Mali: Public Sector Debt Sustainability Framework, Baseline Scenario, 2017–2040

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
Public sector debt 1/	**35.5**	**36.1**	**40.5**	**44.1**	**46.2**	**46.9**	**47.0**	**46.6**	**46.1**	**43.2**	**36.5**	**30.7**	**45.3**
of which: external debt	**24.5**	**23.4**	**26.4**	**26.3**	**25.6**	**25.6**	**25.8**	**26.0**	**26.6**	**28.5**	**28.2**	**22.8**	**26.8**
Change in public sector debt	-0.5	0.6	4.4	3.7	2.1	0.7	0.1	-0.4	-0.5	-0.4	-0.4		
Identified debt-creating flows	**-2.3**	**3.4**	**-0.2**	**3.6**	**2.0**	**0.8**	**0.2**	**-0.2**	**-0.2**	**-0.1**	**-0.4**	**0.9**	**0.5**
Primary deficit	**2.0**	**3.9**	**0.7**	**4.3**	**4.2**	**3.0**	**1.9**	**1.4**	**1.4**	**1.4**	**1.4**	**2.1**	**2.1**
Revenue and grants	20.1	15.6	21.4	20.0	21.6	21.2	21.5	21.3	21.3	21.4	16.2	**17.8**	**21.2**
of which: grants	1.6	1.2	1.9	1.4	1.9	1.6	1.8	1.5	1.4	1.2	0.8		
Primary (noninterest) expenditure	22.1	19.4	22.1	24.3	25.8	24.1	23.4	22.7	22.7	22.8	17.6	**19.9**	**23.4**
Automatic debt dynamics	**-4.2**	**-0.4**	**-0.9**	**-0.4**	**-1.3**	**-2.1**	**-1.6**	**-1.6**	**-1.6**	**-1.6**	**-1.8**		
Contribution from interest rate/growth differential	-1.7	-1.5	-1.3	1.4	-1.0	-1.9	-1.5	-1.5	-1.6	-1.6	-1.7		
of which: contribution from average real interest rate	0.2	0.1	0.3	0.6	0.7	0.7	0.8	0.7	0.6	0.5	0.1		
of which: contribution from real GDP growth	-1.8	-1.6	-1.6	0.8	-1.7	-2.6	-2.2	-2.2	-2.2	-2.1	-1.8		
Contribution from real exchange rate depreciation	-2.6	1.1	0.5	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**-0.1**	**-0.1**	**0.0**	**-0.3**	**-0.9**	**-0.1**	**-0.1**	**-0.1**	**0.0**	**0.0**	**0.0**	**-0.2**	**-0.1**
Privatization receipts (negative)	0.1	0.1	0.1	0.0	-0.6	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)*	-0.2	-0.2	-0.2	-0.3	-0.2	-0.1	-0.1	-0.1	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**1.8**	**-2.7**	**4.6**	**-1.8**	**-0.3**	**-0.3**	**-0.3**	**-0.2**	**-0.3**	**-0.2**	**-0.1**	**0.9**	**-0.5**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**34.7**	**38.4**	**40.5**	**40.9**	**40.8**	**40.1**	**39.5**	**36.1**	**30.1**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**162.0**	**192.2**	**187.4**	**193.4**	**189.6**	**188.5**	**185.4**	**168.6**	**185.6**		
Debt service-to-revenue and grants ratio 3/	**31.8**	**32.5**	**23.8**	**31.7**	**33.7**	**49.5**	**46.4**	**48.5**	**42.6**	**35.9**	**30.7**		
Gross financing need 4/	8.3	8.8	5.7	10.5	10.7	13.4	11.8	11.7	10.5	9.1	6.4		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	5.3	4.7	4.8	-2.0	4.0	6.0	5.0	5.0	5.0	5.0	5.0	**4.4**	**4.4**
Average nominal interest rate on external debt (in percent)	1.2	1.0	1.4	1.5	1.5	1.5	1.6	1.6	1.2	1.5	1.7	**0.6**	**1.4**
Average real interest rate on domestic debt (in percent)	3.1	4.0	3.1	3.7	4.5	4.0	4.1	4.0	4.3	4.4	3.3	**3.0**	**4.2**
Real exchange rate depreciation (in percent, + indicates depreciation)	-11.0	4.9	2.1	...	...	...	...	...	...	...	...	**1.5**	**...**
Inflation rate (GDP deflator, in percent)	1.9	1.5	2.1	2.0	1.5	2.0	2.0	2.0	2.0	2.0	2.7	**3.3**	**2.0**
Growth of real primary spending (deflated by GDP deflator, in percent)	7.8	-7.9	19.1	7.8	10.5	-0.8	1.8	1.8	5.1	5.1	-19.5	**5.3**	**4.7**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	2.5	3.2	-3.7	0.7	2.1	2.3	1.8	1.8	1.9	1.8	1.8	**0.7**	**1.9**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/



Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, government-guaranteed debt . Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

* CCRT-related debt relief is included in the revenue as *a capital grant* in the fiscal account, and thereby included in the primary deficit but not in the "debt relief (HIPC and other)".

Figure 1. Mali: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2020–30

Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	n.a.	n.a.
Commodity Price 2/	No	No
Market financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.6%	1.6%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	24	24
Avg. grace period	5	5

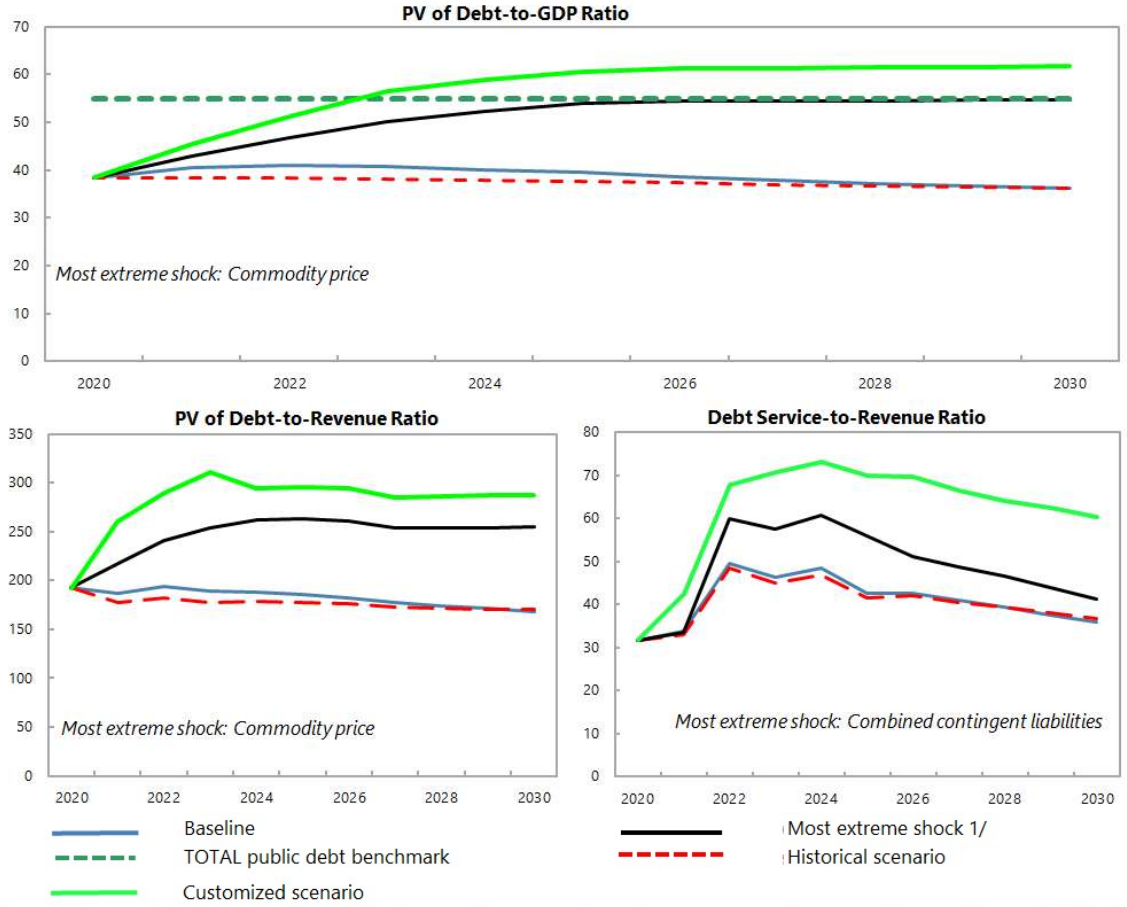
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Mali: Indicators of Public Debt Under Alternative Scenarios, 2020–30



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	32%	32%
Domestic medium and long-term	68%	68%
Domestic short-term	0%	0%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.6%	1.6%
Avg. maturity (incl. grace period)	24	24
Avg. grace period	5	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	3.9%	3.9%
Avg. maturity (incl. grace period)	4	4
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	4.5%	4.5%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Mali: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2020–2030

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of debt-to GDP ratio											
Baseline	21	20	20	19	19	20	20	20	21	21	21
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	21	22	23	24	24	24	23	22	20	19	16
Customized scenario: larger fiscal deficits financed on non-concessional terms with export growth shocks	21	24	28	29	27	27	26	25	24	24	24
B. Bound Tests											
B1. Real GDP growth	21	21	21	21	21	22	22	22	22	23	23
B2. Primary balance	21	20	20	20	20	21	21	21	21	22	22
B3. Exports	21	21	23	23	23	24	24	24	24	24	24
B4. Other flows 3/	21	23	25	25	24	25	25	25	25	25	25
B5. Depreciation	21	25	21	21	21	22	22	23	23	24	24
B6. Combination of B1-B5	21	24	24	24	24	25	25	25	25	25	25
C. Tailored Tests											
C1. Combined contingent liabilities	21	22	22	23	23	24	24	25	25	26	26
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	21	22	23	23	23	23	23	23	22	22	22
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	79	76	79	82	86	93	97	102	107	113	119
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	79	84	93	101	105	110	111	110	106	100	91
Customized scenario: larger fiscal deficits financed on non-concessional terms with export growth shocks	79	100	128	138	139	142	142	143	145	148	154
B. Bound Tests											
B1. Real GDP growth	79	76	79	82	86	93	97	102	107	113	119
B2. Primary balance	79	77	80	84	89	96	101	106	111	117	123
B3. Exports	79	89	108	112	117	125	131	136	141	146	152
B4. Other flows 3/	79	86	100	104	108	115	120	124	128	132	137
B5. Depreciation	79	76	68	72	75	82	86	91	97	103	109
B6. Combination of B1-B5	79	89	89	99	103	110	115	120	125	130	135
C. Tailored Tests											
C1. Combined contingent liabilities	79	85	89	95	101	111	117	124	130	137	144
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	79	90	101	104	107	112	113	116	118	120	123
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	7	7	7	8	8	6	7	8	8	8	9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	7	7	8	9	9	7	9	9	10	10	10
Customized scenario: larger fiscal deficits financed on non-concessional terms with export growth shocks	7	7	9	13	15	14	15	15	14	13	12
B. Bound Tests											
B1. Real GDP growth	7	7	7	8	8	6	7	8	8	8	9
B2. Primary balance	7	7	7	8	8	6	7	8	8	9	9
B3. Exports	7	7	9	9	10	7	9	10	11	11	11
B4. Other flows 3/	7	7	8	8	9	6	8	9	10	10	10
B5. Depreciation	7	7	7	7	8	6	7	8	7	8	8
B6. Combination of B1-B5	7	7	8	8	9	7	8	9	10	10	10
C. Tailored Tests											
C1. Combined contingent liabilities	7	7	7	8	8	6	8	8	9	9	9
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	7	7	8	9	9	7	8	9	10	10	10
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	10	9	9	9	9	7	8	8	8	8	8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	10	9	10	10	11	8	9	9	10	9	9
Customized scenario: larger fiscal deficits financed on non-concessional terms with export growth shocks	10	10	12	15	15	13	14	13	12	10	9
B. Bound Tests	10	10	12	15	15	13	14	13	12	10	9
B1. Real GDP growth	10	9	10	10	10	7	8	8	8	8	8
B2. Primary balance	10	9	9	9	9	7	8	8	8	8	8
B3. Exports	10	9	10	10	10	7	8	9	9	9	9
B4. Other flows 3/	10	9	10	10	10	7	8	9	10	9	9
B5. Depreciation	10	11	12	11	11	8	9	9	9	9	9
B6. Combination of B1-B5	10	10	11	10	10	7	9	10	10	9	9
C. Tailored Tests											
C1. Combined contingent liabilities	10	9	10	9	10	7	8	8	8	8	8
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	10	10	11	11	11	8	9	9	9	9	9
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

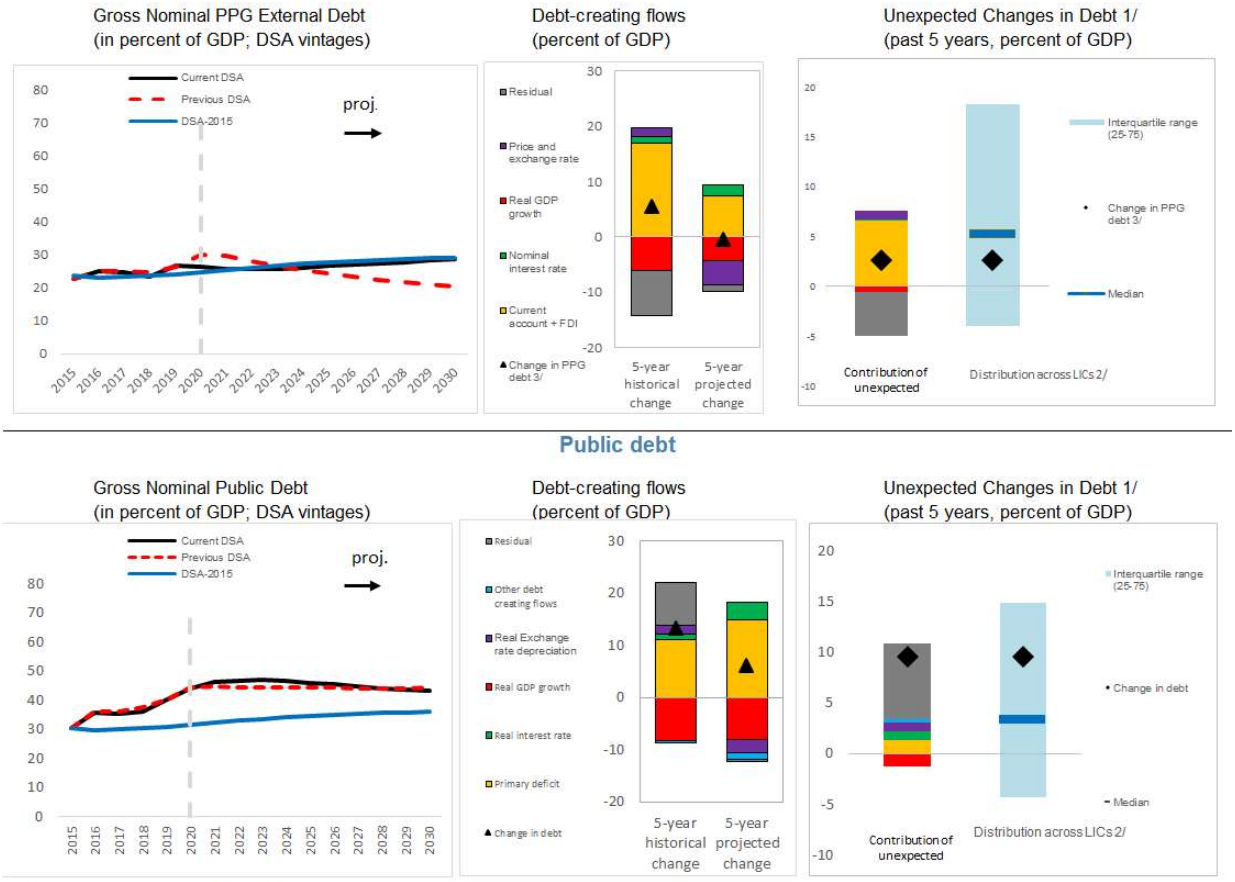
Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Mali: Sensitivity Analysis for Key Indicators of Public Debt, 2020–2030

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of Debt-to-GDP Ratio											
Baseline	38	40	41	41	40	39	39	38	37	37	36
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	38	38	38	38	38	38	37	37	37	36	36
Customized scenario: larger fiscal deficits financed on non-concessional terms with commodity price shocks	38	45	51	**56**	**59**	**61**	**61**	**61**	**61**	**62**	**62**
B. Bound Tests											
B1. Real GDP growth	38	43	46	47	47	48	48	48	48	49	49
B2. Primary balance	38	41	43	43	42	41	40	39	39	38	37
B3. Exports	38	42	45	44	43	43	42	41	40	39	38
B4. Other flows 3/	38	43	46	46	45	44	43	42	41	40	39
B5. Depreciation	38	44	42	41	38	37	34	32	30	29	27
B6. Combination of B1-B5	38	40	41	40	39	38	38	37	36	35	35
C. Tailored Tests											
C1. Combined contingent liabilities	38	51	50	50	49	48	47	46	45	44	43
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	38	43	47	50	52	54	54	54	54	55	55
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
TOTAL public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	192	187	193	189	188	185	182	177	174	171	168
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	192	178	182	178	179	177	176	173	172	171	170
Customized scenario: larger fiscal deficits financed on non-concessional terms with commodity price shocks	32	42	68	71	73	70	69	66	64	62	60
B. Bound Tests											
B1. Real GDP growth	192	197	216	217	222	224	225	224	226	227	229
B2. Primary balance	192	192	202	198	197	193	189	184	180	177	175
B3. Exports	192	194	210	206	204	201	197	191	187	183	179
B4. Other flows 3/	192	200	218	213	211	208	204	197	192	188	184
B5. Depreciation	192	206	202	190	182	172	163	152	143	135	128
B6. Combination of B1-B5	192	185	193	185	184	181	177	172	169	166	164
C. Tailored Tests											
C1. Combined contingent liabilities	192	234	238	232	229	224	220	213	208	204	200
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	192	217	241	254	262	263	261	253	253	254	255
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	32	34	49	46	48	43	43	41	39	38	36
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	32	33	48	45	47	42	42	40	39	38	37
Customized scenario: larger fiscal deficits financed on non-concessional term	32	42	68	71	73	70	69	66	64	62	60
B. Bound Tests											
B1. Real GDP growth	32	35	53	51	55	51	52	51	51	50	49
B2. Primary balance	32	34	50	48	50	45	45	42	41	39	37
B3. Exports	32	34	50	47	49	43	43	42	40	39	37
B4. Other flows 3/	32	34	50	47	49	43	43	42	41	39	37
B5. Depreciation	32	33	49	45	47	41	41	40	38	36	34
B6. Combination of B1-B5	32	33	48	45	47	42	42	40	39	37	35
C. Tailored Tests											
C1. Combined contingent liabilities	32	34	60	58	61	56	51	49	46	44	41
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	32	37	56	55	60	56	58	57	57	56	54
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

Figure 3. Mali: Drivers of Debt Dynamics – Baseline Scenario



1/ Difference between anticipated and actual contributions on debt ratios.

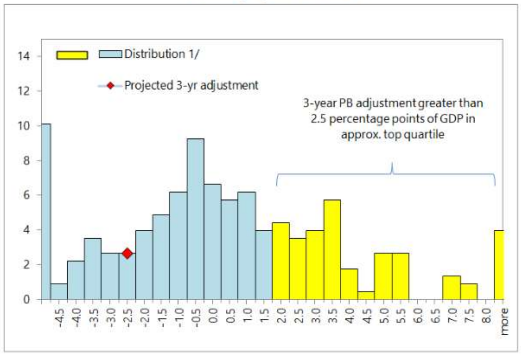
2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

Sources: Malian authorities and staff estimates and projections.

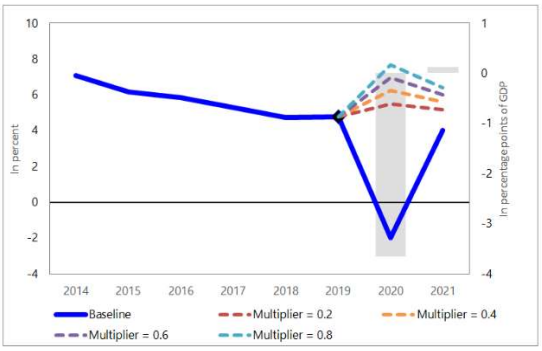
Figure 4. Mali: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



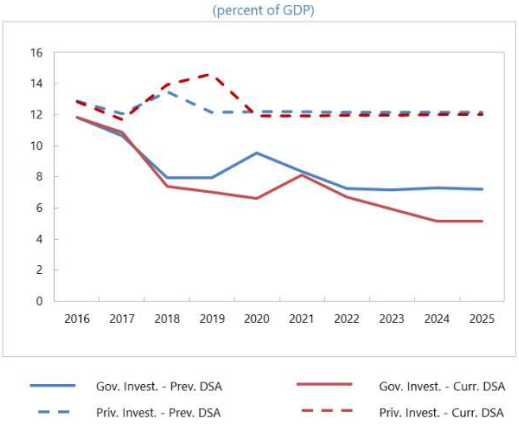
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/

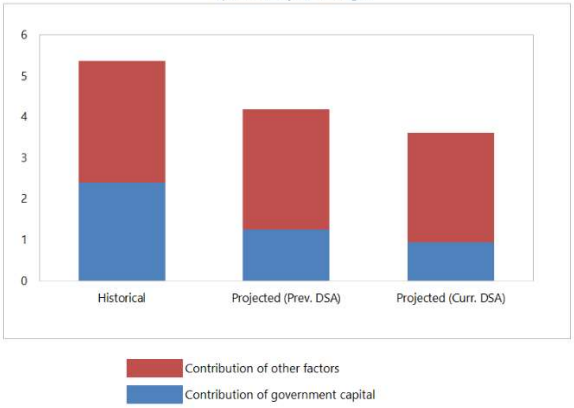


1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)

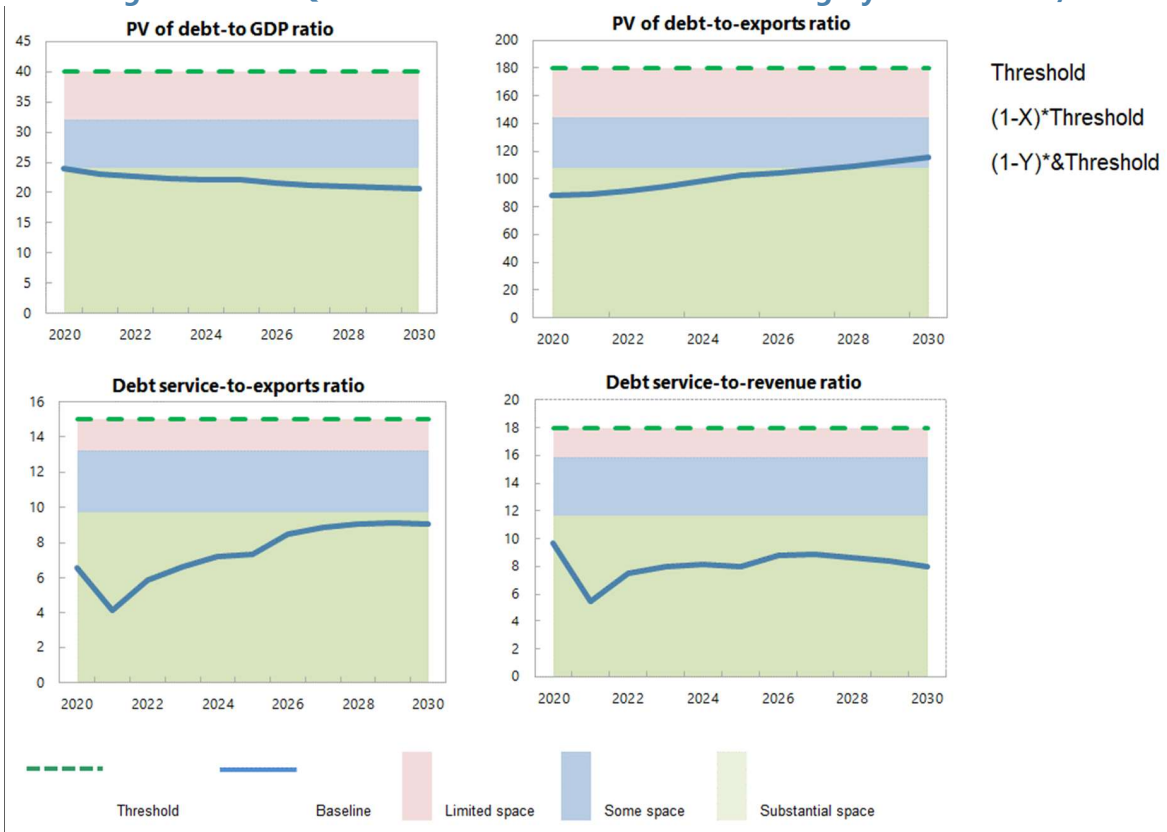


Contribution to Real GDP growth
(percent, 5-year average)



Sources: Malian authorities and staff estimates and projections.

Figure 5. Mali: Qualification of the Moderate Risk Category. 2020–2030 1/



Sources: Country authorities; and staff estimates and projections.
1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.